

TinyDCS three-layer architecture

Layer 1 — Published mechanistic models

ADRAC
(log-logistic AFT)

Conkin RM/NM
(Eq. 14/15)

Gerth 3RUT-MBe1
(bubble ODE)
[calibration WIP]

Layer 2 — TinyDCS surrogate (this paper)

trains on

LightGBM · 13 features · Smithson-Verkuilen target · logit transform
ZERO-INFLATED 2-STAGE: $P(y=0)$ classifier × continuous regressor on $y>0$
Split-conformal on logit residuals · Mahalanobis OOD abstention

Layer 3 — Personalization + fusion (Paper 2)

refines

Hierarchical Bayesian
per-subject log λ
(conjugate Gaussian)

Wearable multimodal fusion
HR, HRV, SpO₂, skin T
(scaffolding only)